

Scientific Computing 2026

Joint distributions, dependence, and random vectors

Lecture 3

Two responses from the same user

One user has click probability $P = 0.1$ or 0.9 , equally likely.
Given P , the responses $X, Y \in \{0, 1\}$ are independent.

Definition: Joint probability mass function

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$$

	$Y = 0$	$Y = 1$	Total
$X = 0$	0.41	0.09	0.50
$X = 1$	0.09	0.41	0.50
Total	0.50	0.50	1

$$p_{X,Y}(1, 1) = \frac{1}{2}(0.1)^2 + \frac{1}{2}(0.9)^2 = 0.41.$$

Marginals discard the pairing

Definition: Marginal PMFs

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Same user

$$(p_{X,Y}(x, y)) = \begin{pmatrix} 0.41 & 0.09 \\ 0.09 & 0.41 \end{pmatrix}$$

Agreement is more likely.

Both models have $X, Y \sim \text{Bernoulli}(0.5)$. The marginals do not determine the joint distribution.

Independently selected users

$$(p_{X,Y}(x, y)) = \begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{pmatrix}$$

All pairs are equally likely.

A conditional PMF normalizes one row

For the same user,

$$\mathbb{P}(Y = 1 \mid X = 1) = \frac{0.41}{0.50} = 0.82, \quad \mathbb{P}(Y = 1 \mid X = 0) = \frac{0.09}{0.50} = 0.18.$$

Definition: Conditional PMF

For $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

Rearranging gives the factorization

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y \mid x).$$

Independence depends on what we condition on

Independence requires

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y.$$

The shared-user table fails: $0.41 \neq 0.5 \cdot 0.5$.

Definition: Conditional independence given P

Within each user type, the joint PMF factors:

$$p_{X,Y|P}(x, y \mid p) = p_{X|P}(x \mid p) p_{Y|P}(y \mid p).$$

The first click provides information about the user's probability P . This does not mean that one click causes the other.

Joint densities assign probability to regions

Near (x, y) , a small rectangle has probability approximately

$$f_{X,Y}(x, y) \Delta x \Delta y.$$

Definition: Joint density

A nonnegative function with total integral one, satisfying

$$\mathbb{P}((X, Y) \in R) = \iint_R f_{X,Y}(x, y) dx dy$$

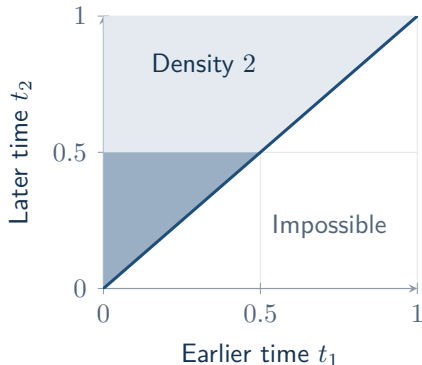
for regions R in the plane.

Integrating over a region adds the probabilities of its small rectangles.

Two arrivals, recorded in order

Independent $U, V \sim \text{Uniform}(0, 1)$, in hours.

Record $T_1 = \min(U, V)$ and $T_2 = \max(U, V)$.



Either person can arrive first:

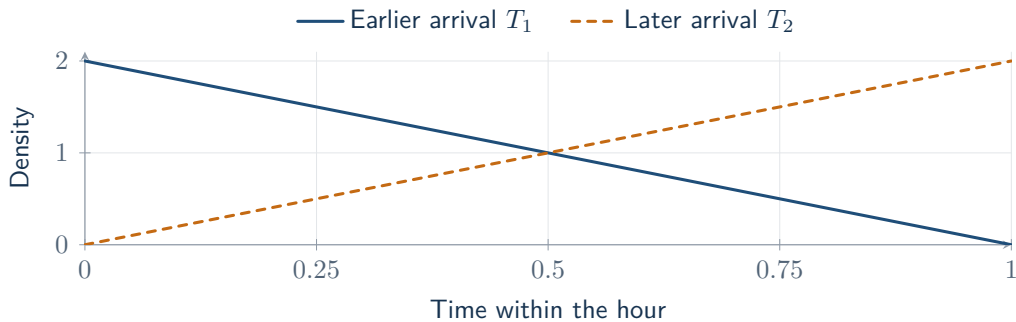
$$f_{T_1, T_2}(t, s) = 2, \quad 0 < t < s < 1.$$

Density is zero elsewhere. Both arrive before half an hour:

$$\mathbb{P}(T_2 \leq \tfrac{1}{2}) = 2 \cdot \tfrac{1}{8} = \tfrac{1}{4}.$$

Sorting introduces dependence: the later arrival must follow the earlier one.

Marginals integrate over the other arrival



$$f_{T_1}(t) = \int_t^1 2 \, ds = 2(1 - t), \quad f_{T_2}(s) = \int_0^s 2 \, dt = 2s.$$

An early first arrival leaves more possible times for the second. In general,

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y) \, dy.$$

A conditional density normalizes one slice

Given $T_1 = t$, the later arrival can be anywhere in $(t, 1)$.

Definition: Conditional density

Where $f_X(x) > 0$,

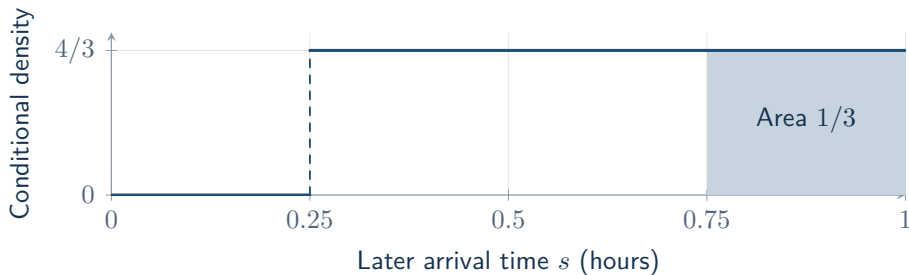
$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

$$f_{T_2|T_1}(s | t) = \frac{2}{2(1-t)} = \frac{1}{1-t}, \quad t < s < 1.$$

For smooth densities, this is the limit of conditioning on a small interval around x . The denominator is a density.

Conditional probability of a late second arrival

The first arrival is at $t = \frac{1}{4}$. The second is uniform over the remaining $\frac{3}{4}$ hour.



$$\mathbb{P}(T_2 > \frac{3}{4} \mid T_1 = \frac{1}{4}) = \int_{3/4}^1 \frac{4}{3} ds = \frac{1}{3}.$$

The shaded area is the probability that the second arrival occurs in the final quarter-hour, given the first arrival at 15 minutes.

The expected gap between arrivals

The expectation rule now uses the joint density:

$$\mathbb{E}[g(X, Y)] = \iint_{\mathbb{R}^2} g(x, y) f_{X,Y}(x, y) dx dy.$$

For the gap $g(t, s) = s - t$,

$$\begin{aligned}\mathbb{E}[T_2 - T_1] &= \int_0^1 \int_t^1 (s - t) 2 ds dt \\ &= \int_0^1 (1 - t)^2 dt = \frac{1}{3} \text{ hour}.\end{aligned}$$

For a discrete pair, replace the integrals by sums. A finite expectation requires $\mathbb{E}[|g(X, Y)|] < \infty$.

Correlation removes the units

For the click table,

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.41 - 0.25 = 0.16.$$

Definition: Correlation

For finite, positive standard deviations σ_X, σ_Y ,

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \mathbb{E}\left[\frac{X - \mu_X}{\sigma_X} \frac{Y - \mu_Y}{\sigma_Y}\right].$$

For the clicks, $\sigma_X = \sigma_Y = 0.5$, so $\rho_{X,Y} = 0.64$.

Changing minutes to seconds changes covariance but leaves correlation unchanged.

Why correlation lies between -1 and 1

Standardize: $U = (X - \mu_X)/\sigma_X$, $V = (Y - \mu_Y)/\sigma_Y$.

$$0 \leq \mathbb{E}[(U - V)^2] = 2 - 2\rho, \quad 0 \leq \mathbb{E}[(U + V)^2] = 2 + 2\rho.$$

Key result: Correlation bound and equality cases

For finite, positive marginal variances, $-1 \leq \rho \leq 1$.

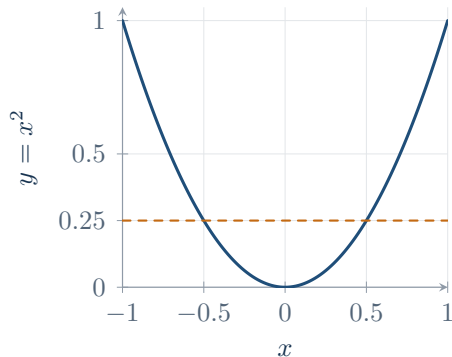
$$\rho = 1 \iff U = V \text{ almost surely}, \quad \rho = -1 \iff U = -V \text{ almost surely.}$$

Perfect correlation means an exact affine relationship.

If either variable is constant, correlation is undefined.

Zero correlation can hide complete dependence

Let $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$.



Symmetry gives

$$\mathbb{E}[X] = \mathbb{E}[X^3] = 0.$$

Therefore

$$\text{Cov}(X, Y) = 0, \quad \rho_{X,Y} = 0.$$

Yet knowing X determines Y exactly.

$\mathbb{P}(Y \leq 1/4 \mid |X| \leq 1/2) = 1$, while $\mathbb{P}(Y \leq 1/4) = 1/2$. Independence fails.

Conditional means minimize squared prediction error

Use X to predict Y . Write

$$m(x) = \mathbb{E}[Y \mid X = x], \quad v(x) = \text{Var}(Y \mid X = x).$$

Theorem: Best prediction under squared error

If $\mathbb{E}[Y^2] < \infty$, every rule $a(X)$ with finite mean squared error satisfies

$$\mathbb{E}[(Y - a(X))^2] = \mathbb{E}[v(X)] + \mathbb{E}[(m(X) - a(X))^2].$$

The minimum is $\mathbb{E}[v(X)]$, attained by $a(X) = m(X)$.

The two terms are remaining uncertainty and the extra error from using a different prediction.

Why the conditional mean is optimal

Inside the group $X = x$, split the prediction error:

$$Y - a(x) = (Y - m(x)) + (m(x) - a(x)).$$

The cross term averages to zero because

$$\mathbb{E}[Y - m(x) \mid X = x] = 0.$$

Squaring and taking conditional expectation gives

$$\mathbb{E}[(Y - a(x))^2 \mid X = x] = v(x) + (m(x) - a(x))^2.$$

Total expectation averages this identity over X .

Prediction in the click model

Without the first response, the best constant prediction is 0.5.

$$\mathbb{E}[(Y - 0.5)^2] = 0.25.$$

After observing it,

$$m(0) = 0.18, \quad m(1) = 0.82.$$

Both groups have the same conditional variance:

$$\mathbb{E}[(Y - m(X))^2] = 0.18(0.82) = 0.1476.$$

These are predictions from a specified probability model. Learning the rule from data is a separate estimation problem.

Several measurements on one object

Record several sensor readings, or several features of one observation.

Definition: Random vector and mean vector

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_d \end{pmatrix} \in \mathbb{R}^d, \quad \boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix}$$

All coordinates belong to the same object or experimental repetition.
A realized observation is a lowercase vector $\mathbf{x}$.

The joint distribution assigns probabilities to regions in $\mathbb{R}^d$.

A covariance matrix records every pair

Definition: Covariance matrix

For a random vector with finite second moments,

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top], \quad \Sigma_{ij} = \text{Cov}(X_i, X_j).$$

Diagonal entries are variances. Off-diagonal entries are covariances.

For the two clicks,

$$\boldsymbol{\mu} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.25 & 0.16 \\ 0.16 & 0.25 \end{pmatrix}.$$

Standardizing the coordinates gives $R_{ij} = \Sigma_{ij} / \sqrt{\Sigma_{ii}\Sigma_{jj}}$. Here R has diagonal 1 and off-diagonal 0.64.

Every weighted sum must have nonnegative variance

For $S = \mathbf{a}^\top \mathbf{X} = \sum_i a_i X_i$,

$$\text{Var}(S) = \sum_i \sum_j a_i a_j \Sigma_{ij} = \mathbf{a}^\top \Sigma \mathbf{a} \geq 0.$$

Definition: Positive semidefinite matrix

A symmetric matrix Σ is positive semidefinite when

$$\mathbf{a}^\top \Sigma \mathbf{a} \geq 0 \quad \text{for every real } \mathbf{a}.$$

It is positive definite when this is strictly positive for every nonzero $\mathbf{a}$.

Every covariance matrix is positive semidefinite. A zero-variance combination can make it singular.

Positive diagonal entries are not enough

Could this be a covariance matrix?

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

It would imply

$$\text{Var}(X_1 - X_2) = 1 + 1 - 2(2) = -2.$$

Key result: Valid covariance matrices in two dimensions

$$\begin{pmatrix} v_1 & c \\ c & v_2 \end{pmatrix} \succeq 0 \iff v_1 \geq 0, \quad v_2 \geq 0, \quad c^2 \leq v_1 v_2$$

With positive variances, the last condition is exactly $|\rho| \leq 1$. A diagonal covariance matrix alone does not guarantee independence.

Means and covariances after a linear transformation

Let $\mathbf{W} = A\mathbf{X} + \mathbf{b}$, with fixed A and $\mathbf{b}$.

Theorem: Linear transformation of moments

For $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}$ and finite covariance Σ ,

$$\mathbb{E}[\mathbf{W}] = A\boldsymbol{\mu} + \mathbf{b}, \quad \text{Cov}(\mathbf{W}) = A\Sigma A^\top.$$

Proof. Centering removes the shift:

$$\mathbf{W} - \mathbb{E}[\mathbf{W}] = A(\mathbf{X} - \boldsymbol{\mu}).$$

Take the expected outer product and pull out the fixed matrices.

These rules need neither independence nor a Gaussian distribution.

Average error and disagreement between sensors

Centered errors have $\Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$.

$$M = \frac{X_1 + X_2}{2}, \quad D = X_1 - X_2, \quad A = \begin{pmatrix} 1/2 & 1/2 \\ 1 & -1 \end{pmatrix}.$$

$$\text{Cov} \begin{pmatrix} M \\ D \end{pmatrix} = A \Sigma A^\top = \begin{pmatrix} 0.9 & 0 \\ 0 & 0.4 \end{pmatrix}.$$

Averaging retains much of the shared error.

Taking the difference cancels much of it.

With independent unit-variance errors, the variances would be 0.5 and 2. Zero covariance between M and D does not yet establish independence.

Correlated Gaussian readings from independent noise

Take independent $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ and set

$$X_1 = Z_1, \quad X_2 = 0.8Z_1 + 0.6Z_2.$$

Both sensors contain the disturbance Z_1 .

Only the second contains the additional disturbance Z_2 .

$$\text{Var}(X_2) = 0.8^2 + 0.6^2 = 1, \quad \text{Cov}(X_1, X_2) = 0.8.$$

Both marginals are standard Gaussian. Their joint distribution records the shared noise.

The multivariate Gaussian construction

Definition: Multivariate Gaussian

Let $\mathbf{Z}$ have independent standard Gaussian coordinates. For fixed A and $\boldsymbol{\mu}$,

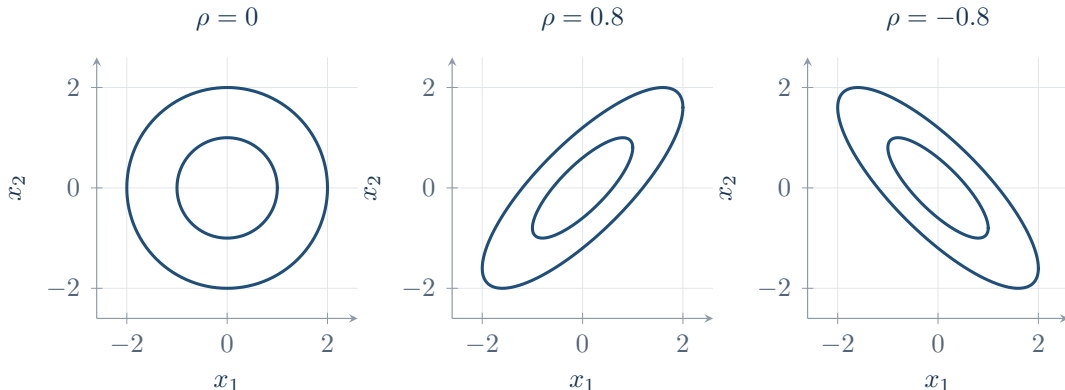
$$\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma), \quad \Sigma = AA^\top.$$

Every linear combination is Gaussian:

$$\mathbf{a}^\top \mathbf{X} \sim \mathcal{N}(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \Sigma \mathbf{a}).$$

Zero variance means a constant. The rows of A specify which independent noises each measurement shares.

Correlation changes the joint shape



In every panel, both marginals are $\mathcal{N}(0, 1)$. Each curve joins points of equal density.
Outer curves have lower density.

The bivariate density from a change of coordinates

For $|\rho| < 1$, write

$$X_1 = Z_1, \quad X_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2.$$

The inverse uses $z_1 = x_1$, $z_2 = (x_2 - \rho x_1)/\sqrt{1 - \rho^2}$. The transformation multiplies areas by $\sqrt{1 - \rho^2}$.

Key result: Standard bivariate Gaussian density

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1 - \rho^2}} \exp\left[-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1 - \rho^2)}\right]$$

Substitute the inverse into the independent-noise density and divide by the area scale to preserve probability.

The long and short directions of a Gaussian ellipse

For $\rho = 0.8$, use rotated coordinates

$$U = \frac{X_1 + X_2}{\sqrt{2}}, \quad V = \frac{X_1 - X_2}{\sqrt{2}}.$$

Their variances are 1.8 and 0.2, with covariance zero.

The exponent depends on

$$q = \frac{u^2}{1.8} + \frac{v^2}{0.2}.$$

At $q = c$, the axis lengths are proportional to $\sqrt{1.8c}$ and $\sqrt{0.2c}$.

The equal-sign direction is three times longer. Shared sensor fluctuations stretch the distribution along the diagonal.

The Gaussian density in matrix notation

Key result: Multivariate Gaussian density

For a positive-definite covariance matrix Σ ,

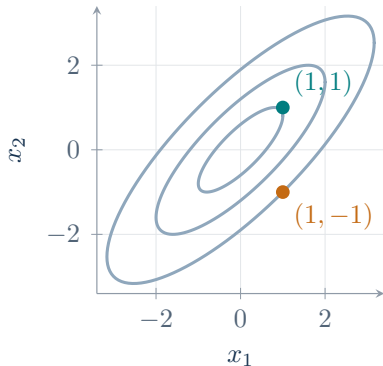
$$f_{\mathbf{x}}(\mathbf{x}) = \frac{\exp[-\frac{1}{2}q(\mathbf{x})]}{(2\pi)^{d/2}\sqrt{\det \Sigma}}, \quad q(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

$q(\mathbf{x})$ is the sum of squared independent-noise coordinates.

$\sqrt{\det \Sigma}$ accounts for the change in volume.

Σ^{-1} is the inverse and $\det \Sigma$ the determinant. For diagonal Σ , the density factors into univariate Gaussian densities.

Equal ordinary distance, different Gaussian density



For unit variances and $\rho = 0.8$,

$$q(1, 1) = \frac{10}{9}, \quad q(1, -1) = 10.$$

Both points are distance $\sqrt{2}$ from the origin. The disagreement $(1, -1)$ has much lower density.

Definition: Squared Mahalanobis distance

$$q(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

Gaussian transformations and independence

Theorem: Closure under linear transformations

If $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, then for fixed $B, \mathbf{b}$,

$$B\mathbf{X} + \mathbf{b} \sim \mathcal{N}(B\boldsymbol{\mu} + \mathbf{b}, B\Sigma B^\top).$$

Selecting coordinates gives Gaussian marginals.

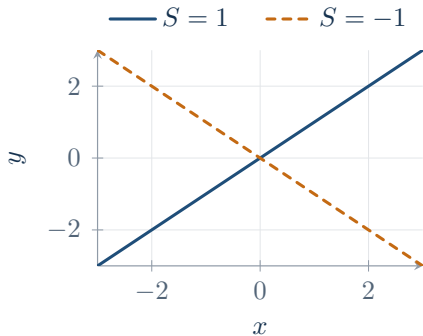
Theorem: The joint Gaussian independence exception

For jointly Gaussian variables, zero covariance implies independence.

Thus the sensor average and difference are independent under the joint Gaussian model.
The moment rules now specify their full distribution.

Gaussian marginals do not guarantee a joint Gaussian

Take $X \sim \mathcal{N}(0, 1)$ and an independent random sign $S = \pm 1$, each with probability $1/2$. Set $Y = SX$.



$Y \sim \mathcal{N}(0, 1)$, and

$$\text{Cov}(X, Y) = \mathbb{E}[SX^2] = 0.$$

But $|Y| = |X|$ always.

The variables are dependent. $X + Y$ equals zero with probability $1/2$, but is not constant.

A joint Gaussian would make every linear combination Gaussian. This one has a point mass, so the pair is not jointly Gaussian.

A singular Gaussian lives on a lower-dimensional set

Let $X_2 = X_1 \sim \mathcal{N}(0, 1)$.

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det \Sigma = 0, \quad \text{Var}(X_1 - X_2) = 0.$$

Every observation lies on the line $x_2 = x_1$.

This is a valid Gaussian random vector, but it has no density with respect to two-dimensional area.

The Gaussian construction allows singular covariance matrices. The full-dimensional density formula requires positive definiteness.

Observing one sensor reveals the shared noise

Recall the construction

$$X_1 = Z_1, \quad X_2 = 0.8Z_1 + 0.6Z_2.$$

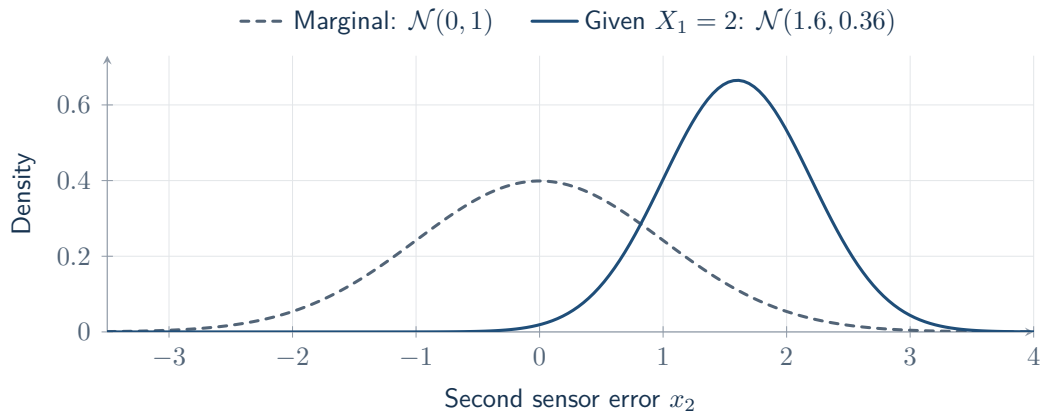
Observing $X_1 = x$ fixes $Z_1 = x$. Only Z_2 remains random.

$$X_2 \mid X_1 = x \sim \mathcal{N}(0.8x, 0.36).$$

If the first sensor reads two units too high, the second has conditional mean 1.6 and conditional standard deviation 0.6.

The marginal distribution of X_2 remains $\mathcal{N}(0, 1)$.

Conditioning shifts the mean and reduces the spread



At $X_1 = 2$, the second sensor's mean shifts from 0 to 1.6. Its variance falls from 1 to 0.36.

The general conditional Gaussian formula

Theorem: Conditioning a bivariate Gaussian

For jointly Gaussian X, Y with $\sigma_X, \sigma_Y > 0$ and $|\rho| < 1$,

$$Y \mid X = x \sim \mathcal{N}\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2)\right).$$

The mean's slope is $\text{Cov}(X, Y) / \text{Var}(X)$.

The conditional variance is constant in x in this model.

For the sensors, total variance gives

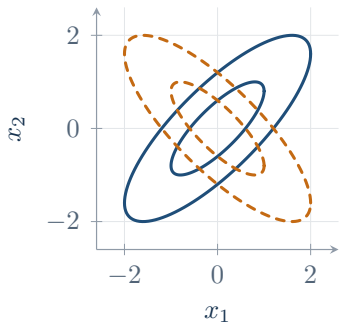
$$\underbrace{1}_{\text{total variance}} = \underbrace{0.36}_{\text{residual variance}} + \underbrace{0.64}_{\text{variance of conditional mean}}.$$

At $|\rho| = 1$, the relationship is exact and the conditional distribution is a point mass.

Two operating modes with identical feature marginals

Two equally likely modes, both with mean zero:

$$\Sigma_+ = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}, \quad \Sigma_- = \begin{pmatrix} 1 & -0.8 \\ -0.8 & 1 \end{pmatrix}.$$



Mode $+$: errors move together.

Mode $-$: errors move in opposite directions. Each feature alone is $\mathcal{N}(0, 1)$ in either class.

The distinguishing information is entirely in the relationship between the features.

Agreement between features identifies the mode

The determinants are equal, so the density normalization cancels:

$$\log \frac{f_+(x_1, x_2)}{f_-(x_1, x_2)} = \frac{2(0.8)}{1 - 0.8^2} x_1 x_2 = \frac{40}{9} x_1 x_2.$$

With equal priors, choose $+$ if $x_1 x_2 > 0$, and $-$ if $x_1 x_2 < 0$.

At $(1, 1)$,

$$\mathbb{P}(C = + \mid \mathbf{X} = (1, 1)) = \frac{1}{1 + e^{-40/9}} \approx 0.9884.$$

At $(1, -1)$, the probability is approximately 0.0116.

Along either coordinate axis the posterior probabilities tie.

An independence assumption can erase the signal

Suppose we replace each class's joint density by the product of its feature marginals.

$$\tilde{f}_+(x_1, x_2) = \phi(x_1)\phi(x_2) = \tilde{f}_-(x_1, x_2),$$

where ϕ is the standard Gaussian density.

Both approximate models now assign the same density to every pair.

$$\tilde{\mathbb{P}}(C = + \mid \mathbf{X} = \mathbf{x}) = \frac{1}{2}.$$

Conditional independence within the class discards all of the useful class information in this example.

The Gaussian classification score

Model $\mathbb{P}(C = k) = \pi_k$ and $\mathbf{X} \mid C = k \sim \mathcal{N}_d(\boldsymbol{\mu}_k, \Sigma_k)$.

Key result: Posterior-maximizing class score

For $\pi_k > 0$ and positive-definite Σ_k , choose the largest

$$s_k(\mathbf{x}) = \log \pi_k - \frac{1}{2} \log \det \Sigma_k - \frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k).$$

The terms account for class prevalence, density spread, and displacement relative to the class covariance.

The omitted $-\frac{d}{2} \log(2\pi)$ is identical for every class.

Shared covariance gives linear decision boundaries

If every class has the same covariance Σ , the quadratic term $-\frac{1}{2}\mathbf{x}^\top \Sigma^{-1} \mathbf{x}$ cancels in comparisons.

The class-dependent score becomes

$$\mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \frac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k.$$

This is linear in the observed features, plus a constant.

With different covariance matrices, quadratic terms generally remain. The two-mode example uses the interaction $x_1 x_2$.

Learning the means, covariances, and class probabilities from data belongs to statistical estimation.

What the joint model lets us compute

Operation	What it answers
Marginalize	What remains if we ignore a measurement?
Condition	What changes after we observe a measurement?
Transform	How do combined measurements vary?
Compare class densities	Which class best explains the whole observation?

Dependence can matter even when each individual feature looks identical across models.

Correlation summarizes linear dependence. A joint distribution can contain much more.